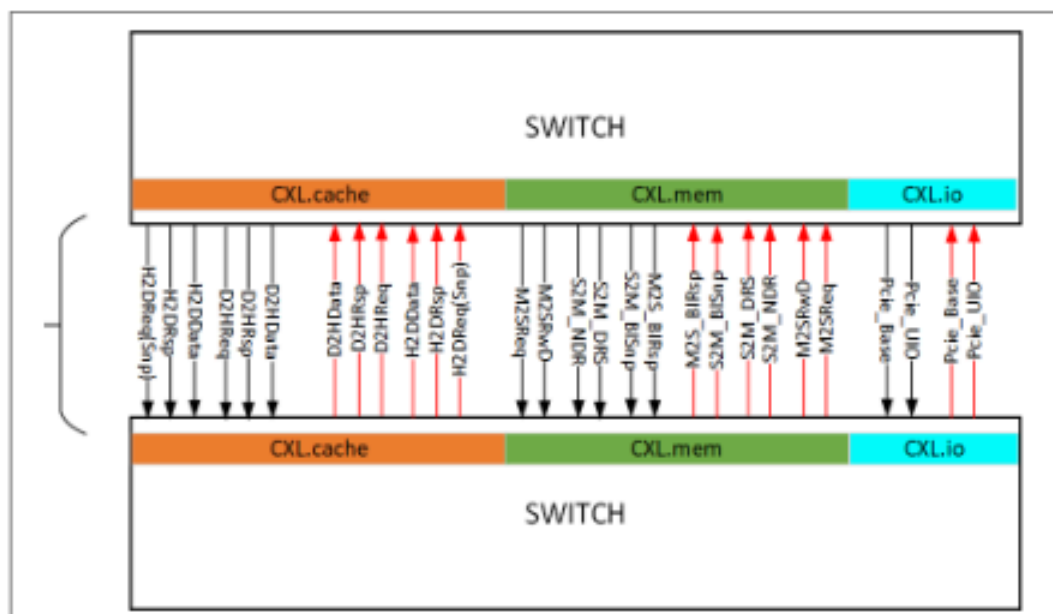


Inter-Switch Links (ISLs)

Inter-Switch Links (ISLs) carry PBR-format flits and must support all message classes and associated sub-channels, including one UIO VC. It is also additionally required that these message classes come up enabled automatically at power on, including the default UIO VC (VC3).

Figure 7-47. ISL Message Class Sub-channels



7.7.7.1 .io Deadlock Avoidance on ISLs/PBR Fabric

ISLs and PBR switches carry CXL.io Upstream traffic and CXL.io Downstream traffic from different hosts in the same physical direction/queues. To avoid deadlocks, these two traffic types need to be kept independent on ISLs and internally through PBR switches. To assist in maintaining the required independence, each TLP inside the PBR fabric is tagged with a DSAR (Downstream Acceptance Rules) bit. Here are the rules for setting the value of the DSAR bit within the PTH:

- When an Edge DSP converts a received TLP from HBR to PBR format, the Edge DSP shall clear the DSAR bit
- When an Edge USP converts a received TLP from HBR to PBR format, the Edge USP shall set the DSAR bit
- When a Host ES vDSP forwards a TLP P2P, it shall set the DSAR bit
- When a GFD sends a TLP (which is always in PBR format), the GFD shall clear the DSAR bit
- When an Edge DSP above a GFD forwards a TLP to the GFD, the Edge DSP shall set the DSAR bit

For the remainder of this section, traffic with DSAR=0 is referred to as USAR (Upstream Acceptance Rules) traffic, and DSAR=1 traffic is referred to as DSAR (Downstream Acceptance Rules) traffic. On an ISL, this bit is carried in the PTH. Traffic within each VC is required to follow the ordering rules specified in [Table 7-107](#) and [Table 7-108](#).

Table 7-107. PBR Fabric .io Ordering Table, Non-UIO

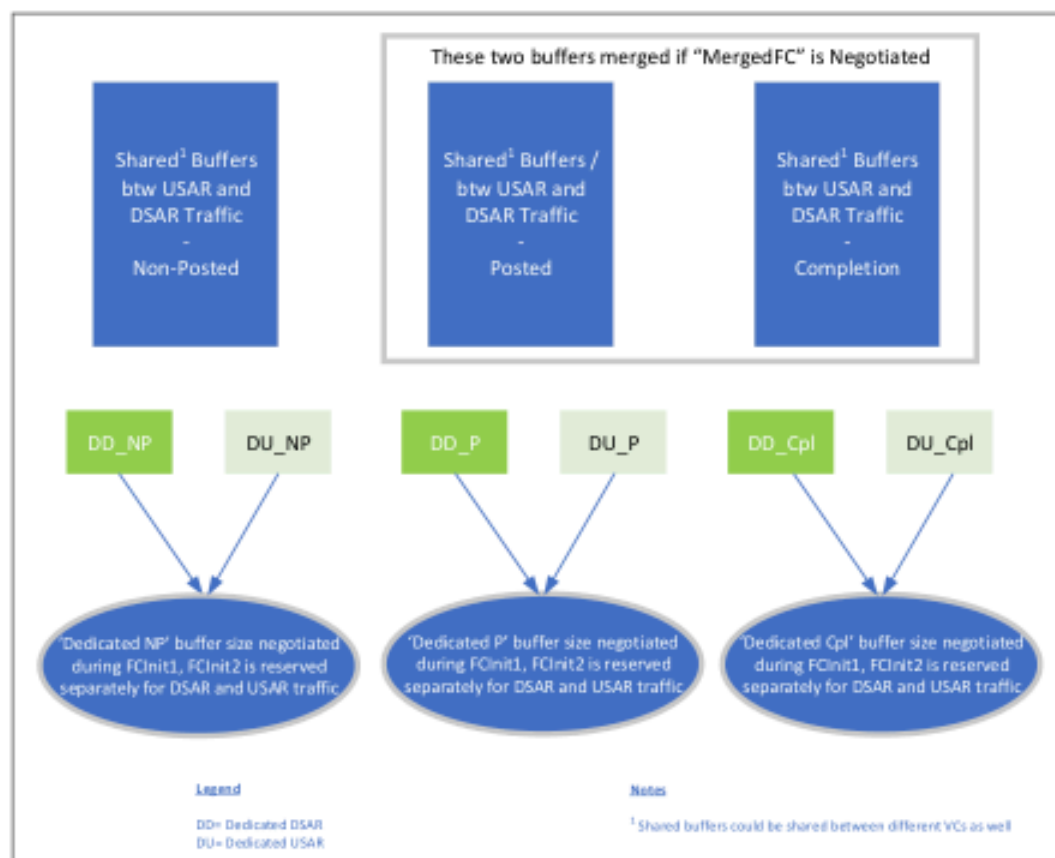
Row Pass Column?			DSAR				USAR			
			Posted Request	Non-Posted Request		Completion	Posted Request	Non-Posted Request		Completion
				Read Request	NP Request with Data			Read Request	NP Request with Data	
DSAR	Posted Request		Per PCIe Base Specification				Yes	Yes	Yes	Yes
	Non-Posted Request	Read Request					Yes/No	Yes	Yes	Yes/No
		NP Request with data					Yes/No	Yes	Yes	Yes/No
	Completion						Yes	Yes	Yes	Yes
USAR	Posted Request		Yes/No	Yes	Yes	Yes/No	Per PCIe Base Specification			
	Non-Posted Request	Read Request	Yes/No	Yes/No	Yes/No	Yes/No				
		NP Request with data	Yes/No	Yes/No	Yes/No	Yes/No				
	Completion		Yes/No	Yes	Yes	Yes/No				

Table 7-108. PBR Fabric .io Ordering Table, UIO

Row Pass Column?		DSAR			USAR		
		UIO PR-FC TLP	UIO NPR-FC TLP	UIO Completion	UIO PR-FC TLP	UIO NPR-FC TLP	UIO Completion
DSAR	UIO PR-FC TLP	Per PCIe Base Specification			Yes	Yes	Yes/No
	UIO NPR-FC TLP				Yes	Yes	Yes/No
	UIO Completion				Yes	Yes	Yes
USAR	UIO PR-FC TLP	Yes/No	Yes/No	Yes/No	Per PCIe Base Specification		
	UIO NPR-FC TLP	Yes/No	Yes/No	Yes/No			
	UIO Completion	Yes	Yes	Yes/No			

To support the additional ordering requirements stated above, the following rules apply on ISL (also pictorially depicted in [Figure 7-48](#)):

Figure 7-48. Deadlock Avoidance Mechanism on ISL



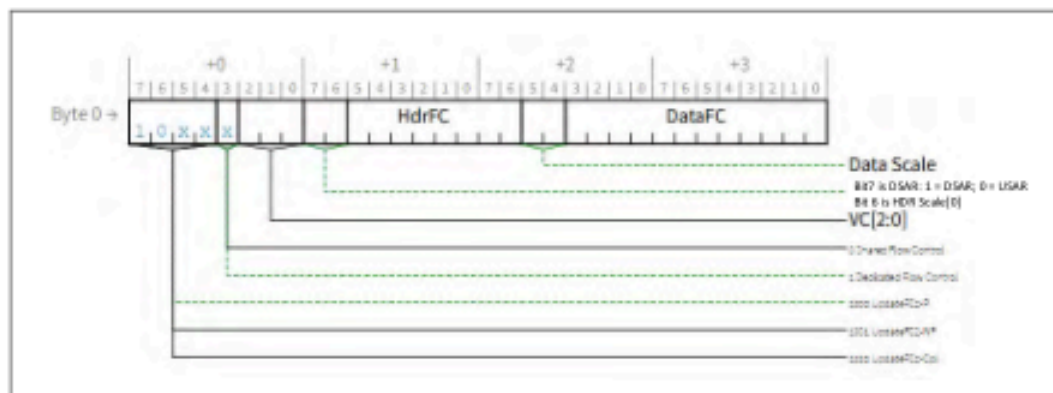
- PBR Fabric .io ordering rules apply independently within each VC implemented
- On edge HBR/PCIe links and on edge PBR links, PBR Fabric ordering rules do not apply
 - On edge PBR links, PTH bit can be ignored for ordering purposes and only the regular CXL.io ordering rules from PCIe Base Specification apply.
- Nonzero dedicated credits are always required on ISL for each VC, regardless of whether multiple VCs are enabled
- Baseline Shared and Merged FC initialization and usage rules, as described in PCIe Base Specification, apply on ISLs as well, with some new rules/exceptions as noted below:
 - Dedicated buffers are required separately per FC class for DSAR and USAR traffic and they are both the same value as negotiated during FC initialization.
 - As an example, if one Posted HDR and one Posted DATA credit were exchanged for Dedicated buffers during InitFC1/2, the transmitter assumes there is 1 Posted data credit for DSAR traffic and one Posted data credit for USAR traffic and similarly for Posted HDR Credit as well.
 - Shared buffers can be shared between DSAR and USAR traffic.
- Update-FC DLLP is modified as shown in Figure 7-49, to indicate release of DSAR or USAR buffers. Transmitters can use this information on shared credits to implement QoS limiting between DSAR and USAR traffic.
 - This modification is implicitly enabled on ISLs and requires no negotiation

Note:

To aid debug, Switches are recommended to capture the Hdr and data_Scale values negotiated at initialization so that debug software can access the values.

- Optimized_Update_FC DLLP applies to USAR traffic only and it is implicit on ISLs. All DSAR traffic's shared buffer credit return occurs only via Update-FC DLLP.

Figure 7-49. Update-FC DLLP Format on ISL



7.7.8 PBR TLP Header (PTH) Rules

For the purposes of this discussion, a “PBR link” is a link that negotiated to PBR flit format via the physical layer TS “PBR Flit bit” (see [Section 6.4](#)). See [Section 3.1.8](#) for details of PTH format.

- A PTH is inserted (via an appropriate decode mechanism) on CXL.io TLPs by an Edge Switch or the PTH is directly generated by devices (e.g., GFD) that natively support PBR link.
- A PTH is forwarded as-is (unless explicitly noted otherwise as in handling PTH.DSAR bit on an edge PBR link) on a CXL.io TLP if the egress port is connected to a PBR link.
- A PTH is removed when its CXL.io TLP exits to an edge non-PBR link.
 - Note that some contents of PTH could be transferred to VendPrefixL0 if the egress port is an HBR link and the VendPrefixL0 is supported and enabled on the link. See [Section 7.7.3](#) and [Section 7.7.4](#).
- A PTH is included in link-IDE Integrity protection, if supported and enabled, when the PTH traverses PBR links.
- PTH is not included in .io selective IDE protection.

7.7.9 PBR Support for UIO Direct P2P to HDM

PBR switches support special routing mechanisms to enable the UIO Direct P2P to HDM use case with edge-to-edge routing, which often can be much more efficient compared to the hierarchical routing used in HBR switches. For backward compatibility, legacy software unaware of these special PBR routing mechanisms can continue to use HDM decoders, providing limited UIO Direct P2P capability.

An enhanced version of the FAST decoder as defined in [Section 7.7.2.4](#) can be implemented in the Edge DSP above the UIO requester, providing edge-to-edge routing for UIO requests that target GFDs.

Another instance of the FAST decoder hardware can provide edge-to-edge routing for UIO requests that target LD-FAM devices. This instance is referred to as an LD-FAM Segment Table (LDST), and it is usually configured with a different segment size and amount of mapped HDM space from any FAST decoders in use.

With LD-FAM devices, UIO completions can be routed edge-to-edge with an ID-Based Re-Router mechanism, which can be implemented in the Edge DSP above each LD-FAM device. The Re-Router matches against the requester's PCI segment number (if applicable) and bus number in the UIO completion to determine the DPID for edge-to-edge routing. G-FAM devices automatically use edge-to-edge routing for UIO completions without this mechanism.

FAST decoders, LDST decoders, and ID-Based Re-Routers are each configured by host software using CCI command sets, as documented in [Section 7.7.14](#) for FAST decoders, and [7.7.13](#) for LDST decoders & ID-based Re-Routers.

GFDs are not associated with any VH, thus they have no PCIe ID (segment, bus, device, function number) assigned by any host. When a GFD sends a UIO completion, the completer segment field (if present) and the completer ID field in the completion are reserved and shall be 0.

7.7.9.1 FAST Decoder Use for UIO Direct P2P to G-FAM

FAST decoder instances in Edge USPs and DSPs have several similarities:

- Both convert requests from HBR format to PBR format, and route edge-to-edge to target GFDs.
- For the SPID, each uses the PID associated with its port.
- Both support CXL.mem and (CXL.io) UIO requests.
- A USP FAST decoder receives HBR format downstream requests coming from the RP. CXL.mem requests result from host accesses to GFDs.
- A DSP FAST decoder receives HBR format upstream requests coming from the requester device. UIO requests result from UIO Direct P2P traffic, where the UIO requester may be directly connected to an Edge DSP, or it may be connected via one or more HBR switches below the Edge DSP. CXL.mem requests result from the Direct P2P CXL.mem for accelerators use case, covered in [Section 7.7.10](#).

A DSP FAST decoder can be configured with a segment size different from the host's USP FAST decoder(s), but it is recommended for all FAST decoders to use the same segment size to avoid software complexity.

A DSP FAST decoder may need to be configured with a different number of segments from the host's USP FAST decoder(s) (e.g., a requester device may not need access to the entire Fabric Address space mapped by the USP FAST decoder). On the other hand, a requester device may need to access the Fabric Address space associated with an entire host Domain, not just a single RP within a host domain.

7.7.9.2 LDST Decoder Use for UIO Direct P2P to LD-FAM

LDST decoder instances in Edge USPs and DSPs have several similarities:

- Both convert requests from HBR format to PBR format, and route edge-to-edge to target LD-FAM devices.
- For the SPID, each uses the PID associated with its port.
- Both support CXL.mem and (CXL.io) UIO requests.
- A USP LDST decoder receives HBR format downstream requests coming from the RP. CXL.mem requests result from host accesses to LD-FAM devices. UIO requests currently have no architected use cases, but they are not prohibited.
- Host software determines whether host accesses to LD-FAM devices use LDST decoders versus HDM Decoders in Edge USPs. For backward compatibility, legacy software that's unaware of LDST decoders can continue to use HDM decoders. For overcoming scaling limitations with the number of HDM decoders supported by

Edge USPs, LDST-aware software can use LDST decoders, though LDST decoders do not support HDM-D.

- A DSP LDST decoder receives HBR format upstream requests coming from the requester device. UIO requests result from UIO Direct P2P traffic. CXL.mem requests result from the Direct P2P CXL.mem for accelerators use case, covered in [Section 7.7.10](#).

A DSP LDST decoder can be configured with a segment size different from the host's USP LDST decoder(s), but it is recommended for all LDST decoders to use the same segment size to avoid software complexity.

A DSP LDST decoder may need to be configured with a different number of segments from the host's USP LDST decoder(s) (e.g., a requester device may not need access to the entire LD-FAM HDM space mapped by the USP LDST decoder). On the other hand, an accelerator may need to access the LD-FAM HDM space associated with the entire host Domain, not a single RP in the host Domain.

When any LDST decoders are in use, host SW needs to configure any HDM decoders mapping the same LD-FAM HDM ranges with decoder characteristics compatible with LDST decoders. This applies to HDM decoders present in the host, PBR switches, HBR switches, or LD-FAM devices. These decoder characteristics include:

- Minimum decoder granularity: 64 GB for LDST
- Interleave Ways (IW): neither HBR nor PBR switches have the special logic required to support 3/6/12, but LDST supports the other architected IW values.

Note that Dynamic Capacity (DC) Block Sizes are not visible to either type of decoder.

LDST decoders insert a requester segment field in UIO requests when necessary. This is described in [Section 7.7.9.3](#).

7.7.9.3 ID-Based Re-Router for UIO Completions with LD-FAM

For UIO Direct P2P to LD-FAM devices, UIO completions by default are routed using hierarchical PCIe ID-based routing, and the ID may include a PCIe segment number in addition to bus, device, and function numbers. If present in the Edge DSP above an LD-FAM device, the ID-Based Re-Router does a CAM match using the PCIe ID, returning the DPID needed for edge-to-edge routing. This mechanism efficiently handles intra-VH cases, and it is especially efficient for cross-VH cases by avoiding P2P through the Root Complex.

PCIe segment numbers in TLPs is a feature added in PCIe Base Specification 6.0, and PCIe segments should not be confused with "segments" in the context of FAST/LDST decoders. LDST decoders support the PCIe convention that requesters generally don't include PCIe segment numbers in requests¹ but rely instead on routing mechanisms to add PCIe segment number fields when needed for cross-segment routing. Host software configures LDST decoders to add² the requester segment field in the request when it targets a different PCIe segment. When the LD-FAM device responds to the UIO request with a UIO completion, it automatically includes segment fields when necessary in the Destination ID and Completer ID. Host software shall configure the ID-Based Re-Router with the PCIe segment number in entries that need it.

1. With Selective IDE non-configuration requests, the requester is required to include the requester segment field in the request because a routing element inserting the field would cause an integrity violation with Selective IDE.
2. Although PCIe Base Specification forbids PCIe switches from inserting a Requester Segment field, the CXL UIO Direct P2P to HDM mechanisms in CXL switches are beyond the scope of PCIe Base Specification and do not violate the underlying architecture principles.

7.7.9.4 LDST and ID-Based Re-Router Access Protection

LDST and ID-Based Re-Router use is protected by the LDST Access Vector (LAV) to ensure that only valid PIDs are programmed by the host into the LDST and ID-Based Re-Router structures. The LAV is a 4k-bit vector with a similar functionality as the GMVs and VTVs.

The FM is responsible for enabling access to PIDs in the LAV before the host can program those PIDs into the LDST or ID-Based Re-Router structures. For cross-VH use cases, the FM is also responsible for using the Domain Validation SV mechanism, when available, to confirm that every VH that is enabled for cross-VH access belongs to the same host domain.

7.7.10 PBR Support for Direct P2P CXL.mem for Accelerators

Direct P2P CXL.mem provides the ability for an accelerator to access peer Type 3 memory devices using CXL.mem. PBR switches require special routing mechanisms to support this, specifically the FAST and LDST decoders. For Direct P2P CXL.mem, these decoders function essentially the same as they do for supporting the UIO Direct P2P or HDM use case, with the following exceptions:

- They intercept and forward upstream CXL.mem requests instead of UIO requests
- They target only Type 3 (HDM) devices, not Type 2 devices
- The accelerator (requester device) and Type 3 device must each be directly connected to an Edge DSP
- With an MLD (Type 3 device), each accelerator must be assigned a dedicated LD distinct from the host's LD

Note that both types of decoders support .mem requests when they are implemented in Edge DSPs, so .mem support is not unique to the Direct P2P CXL.mem use case.

Same as with the UIO Direct P2P use case, a FAST decoder can be implemented in the Edge DSP above an accelerator, providing edge-to-edge routing for .mem requests that target G-FAM devices (GFDs). The same FAST decoder instance can support either the UIO Direct P2P or Direct P2P CXL.mem use case.

Similarly, an LDST decoder can be implemented in the Edge DSP above an accelerator, providing edge-to-edge routing for .mem requests that target LD-FAM devices. The same LDST decoder instance can support either the UIO Direct P2P or Direct P2P CXL.mem use case.

Type 3 devices used with Direct P2P CXL.mem can be mapped under either HDM-H or HDM-DB coherency ranges. If mapped under HDM-DB, peer devices other than the associated accelerator can access the HDM-DB memory using UIO Direct P2P to HDM, in which case the associated accelerator serves the role of the host participating in BI protocol (i.e., the HDM-DB device directs BISnps to the accelerator).

Direct P2P CXL.mem traffic going to or from an MLD (directly connected to an Edge DSP) works essentially the same as with host .mem traffic, as documented in [Section 7.7.6.6.3](#) and [Section 7.7.6.8](#).

CXL.mem responses for the Direct P2P CXL.mem use case require no special routing mechanism. For S2M responses from G-FAM, the GFD's RPID context for the accelerator contains the DPID needed for edge-to-edge routing back to the accelerator. For S2M responses from LD-FAM, the vPPB in the Edge DSP above the Type 3 device contains the DPID needed for edge-to-edge routing back to the accelerator.

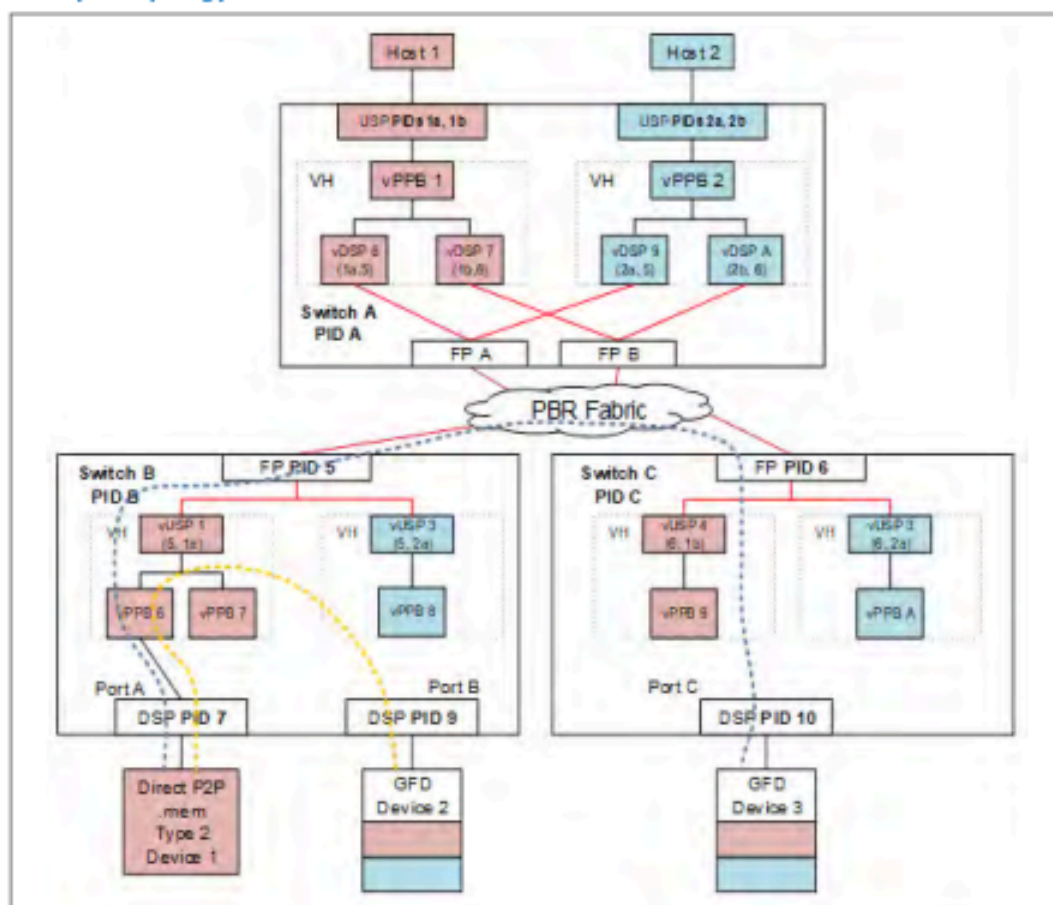
Same as with the UIO Direct P2P use case, FAST decoders and LDST decoders are each configured by host software using CCI command sets, as documented in [Section 7.7.15](#) for FAST decoders and [Section 7.7.13](#) for LDST decoders.

7.7.10.1 Message Routing for Direct P2P CXL.mem Accesses with GFD

Direct P2P CXL.mem messages are routed using standard PBR mechanisms. [Figure 7-50](#) illustrates an example PBR Fabric with a Direct P2P CXL.mem enabled Type 2 accelerator and two peer GFDs accessible to it. The dashed lines indicate the paths taken by the Direct P2P CXL.mem messages. Upstream .mem requests from the accelerator are routed edge-to-edge to the appropriate GFD by the FAST decoder in vPPB 6. Upstream .mem responses from either GFD are routed edge-to-edge back to the accelerator by standard PBR routing.

For an HDM-DB GFD sending a BISnp, the GFD's RPID context for the accelerator contains the DPID that is needed for edge-to-edge routing to the accelerator.

Figure 7-50. Example Topology with Direct P2P CXL.mem with GFD

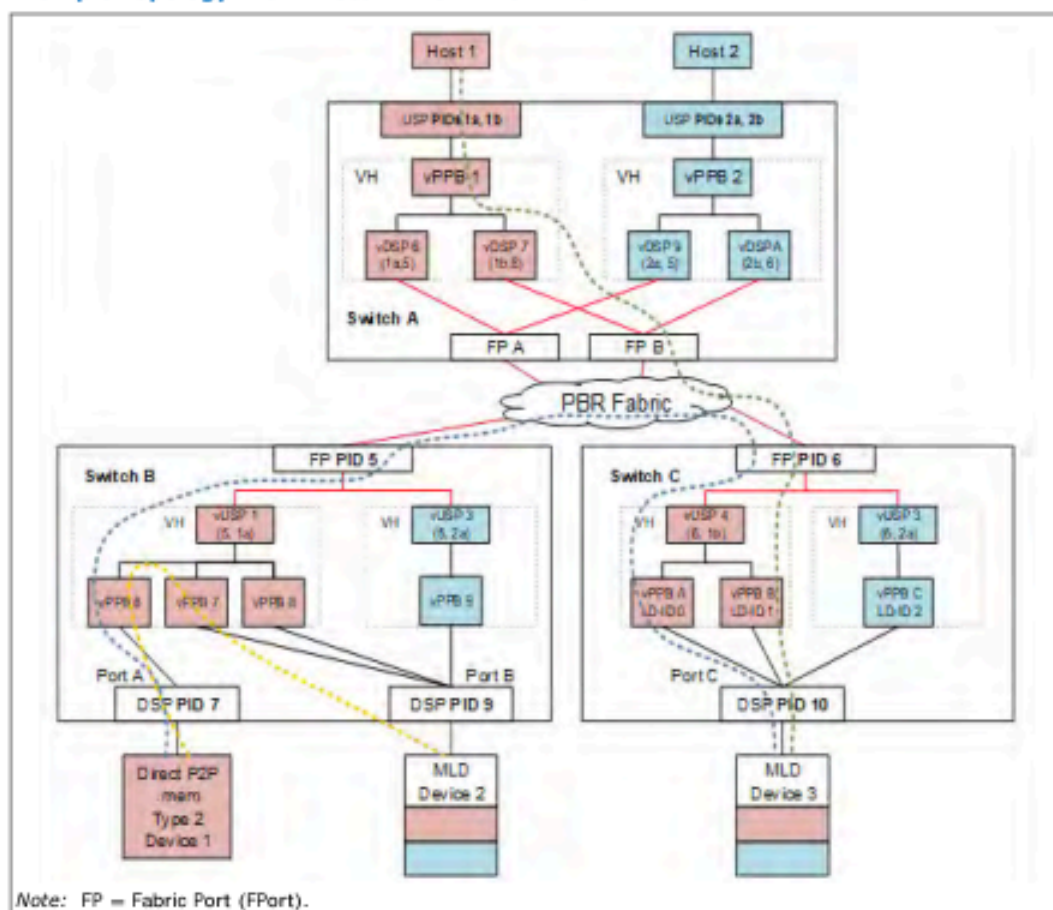


7.7.10.2 Message Routing for Direct P2P CXL.mem Accesses with MLD

Direct P2P CXL.mem accesses to an MLD require a distinct LD and associated peer requester LD-ID on the link between the MLD and the Edge DSP to which it is attached. This is accomplished by assigning a vPPB in the DSP in the same Domain as the host that owns the requester. The host and any peer accelerators will each have their own vPPB bound to them, which utilize their individual LD-IDs.

Figure 7-51 illustrates an example PBR Fabric with a Direct P2P CXL.mem enabled Type 2 accelerator and two peer MLDs accessible to it. Other than the dashed line to Host 1, the dashed lines indicate the paths taken by the Direct P2P CXL.mem messages. Upstream CXL.mem requests from the accelerator are routed edge-to-edge to the appropriate MLD by the LDST decoder in vPPB 6. Upstream CXL.mem responses from either MLD are routed edge-to-edge back to the accelerator by standard PBR routing using the accelerator's PID, which in each case is retrieved from the accelerator's vPPB in the DSP above the MLD.

Figure 7-51. Example Topology with Direct P2P CXL.mem with MLD



In this example, the path taken by CXL.mem messages between the host and one MLD is also shown. Downstream CXL.mem requests from the host are routed edge-to-edge to the appropriate MLD by the LDST decoder in vPPB 1. Upstream CXL.mem responses from the MLD are routed edge-to-edge back to the host by standard PBR routing using the host's PID contained in vPPB B.

For an HDM-DB LD-FAM device sending a BISnp, the Edge DSP above the LD-FAM device contains the DPID that is needed for edge-to-edge routing to the accelerator.

7.7.10.3 PBR Switch Port Processing of Direct P2P CXL.mem Messages

Table 7-109 summarizes how PBR switches perform port processing of CXL.mem messages with the Direct P2P CXL.mem for Accelerators use case. This traffic never flows through Edge USPs or HBR switches. The accelerator (requester) is always an SLD directly connected to an Edge DSP, and each Type 3 memory device is always directly connected to an Edge DSP. All messages in PBR format are routed edge-to-edge.

For conciseness, there are several abbreviations within the table. Beyond “accel” standing for accelerator, see [Section 7.7.6.8](#) for other abbreviations.

Table 7-109. PBR Switch Port Processing Table for Direct P2P CXL.mem

Message Class and Direction		Edge USP Always below RP	Host ES FPort or DS ES FPort	Edge DSP in Either Host ES or Downstream ES			
				Above HBR Switch USP	Above SLD	Above MLD	Above GFD
M2S Req/RwD Direct P2P CXL.mem	US from accel	N/A	PBR routing	N/A	Convert to PBR fmt using FAST or LDST	N/A	N/A
	DS to Type 3				Convert to HBR fmt LD-ID ← 0; is unused	LD-ID ← CAM ₁₆ (SPID) Convert to HBR MLD fmt	LD-ID is N/A Keep in PBR fmt
S2M NDR/DRS Direct P2P CXL.mem	US from Type 3	N/A	PBR routing	N/A	LD-ID is unused Convert to PBR fmt DPID ← vPPB.root.PID	LD-ID identifies vPPB Convert to PBR fmt DPID ← vPPB.root.PID	Keep in PBR fmt LD-ID is N/A
	DS to accel				Convert to HBR fmt LD-ID ← 0; is unused	N/A	N/A
S2M BISnp Direct P2P CXL.mem	US from Type 3	N/A	PBR routing	N/A	Convert to PBR fmt DPID ← vPPB.root.PID SPID ← vPPB.self.PID	BI-ID[3:0] contains LD-ID LD-ID identifies vPPB Convert to PBR fmt DPID ← vPPB.root.PID SPID ← vPPB.self.PID	Keep in PBR fmt
	DS to accel				Convert to HBR fmt BI-ID[11:0] ← SPID	N/A	N/A
M2S BIRsp Direct P2P CXL.mem	US from accel	N/A	PBR routing	N/A	Convert to PBR fmt DPID ← BI-ID[11:0] SPID ← vPPB.self.PID	N/A	N/A
	DS to Type 3				Convert to HBR fmt BI-ID ← 0; is unused	Convert to HBR fmt LD-ID ← CAM ₁₆ (SPID) BI-ID[3:0] ← vPPB.LD-ID	Keep in PBR fmt

7.7.11 PBR Link Events and Messages

A PBR link can carry traffic from many different VH at the same time. Some events may occur that only affect a single VH, while other events need to apply to all VH sharing the link.

Basic PBR link requirements are discussed in [Section 7.7.11.1](#).

A summary of all the CXL Vendor Defined Messages (VDMs) that are PTH routed to the destination is provided in [Section 7.7.11.2](#).

PCIe events for a single VH are discussed in [Section 7.7.11.3](#).

PCIe events for multiple VH sharing a link are discussed in [Section 7.7.11.4](#).

Events that occur outside PCIe are discussed in [Section 7.7.11.5](#).

Messaging to and from a host to a GFD is discussed in [Section 3.1.11.1](#).

7.7.11.1 PBR Link Fundamentals

CXL defines two types of PBR links:

- Inter-Switch Link (ISL)
- GFD link

All PBR links must support PBR Flit mode. Because PBR Flit mode relies on PCIe Flit mode, all host-OS-visible DSPs should report PCIe Flit mode as enabled. The DSPs include both a Host Edge Switch vDSP and a DSP above a PBR link that leads to a GFD.

The owner of a PBR link is an FM-managed DSP. Switch firmware may assist the FM in managing the DSP. An ISL is a downstream-to-downstream crosslink and thus has an FM-managed DSP on each side of the link. A GFD link has only one DSP and thus has only one FM-managed DSP. The speed and width of a PBR link is solely controlled by the FM-managed DSP(s) on the link and not by any vDSPs that share the link.

Each side of an ISL is managed separately. Each DSP above an ISL must support DPC, to allow firmware on each side of the link an independent amount of time to process fabric port events. DPC shall be enabled for all cases on ISL except when the ISL is the only path to the FM, in which case the DSP furthest from the FM shall not have DPC enabled.

FM-initiated CXL.io traffic sent across a PBR link shall be limited to DMTF-format VDMs. The PTH.DPID is used to indicate whether the PBR Link Partner should sink the TLP or forward the TLP. If the PTH.DPID = FFFh, the PBR Link Partner must sink the VDM because that is how the initial device discovery occurs and how PIDs are assigned. If the PTH.DPID = the device's PID, then the device must also sink the VDM because that is how the device is accessed by the FM.

All VH users of a PBR link have their functionality ride on top of the FM-managed link. For example, a VH's DSP cannot see a Link Up if the fabric link is not up. A VH cannot change the width or speed of its shared link, rather it will inherit the setting of the FM-managed DSP.

To manage different software response times to events, every vDSP for every VH must support DPC. DPC allows a host to keep its Link Down from its (VH) perspective until it is ready to re-enable it, having cleaned up all the side effects of a Link Down. A Host may or may not choose to enable DPC.

L0p is optional on a PBR link. The FM-managed DSP initiates any L0p transitions via a mechanism that is beyond this specification.

Every CXL.io TLP on a PBR link will carry a 4B PTH. The VDMs described in this section follow the same rule. See [Section 3.1.8](#). There are three fields of note in the PTH that are required for the VDMs described in this section:

- SPID: Source PID
 - From a vDSP: Use vDSP's USP PID
 - From a vUSP: Use vUSP's FPort PID
 - From a switch: Use switch's PID
 - From a downstream edge: Use DSP's PID
 - From a host edge: Use USP's PID
- DPID: Destination PID
 - To a vDSP: Use vDSP's USP PID
 - To a vUSP: Use vUSP's FPort PID
 - To a switch: Use switch's PID
 - To a downstream edge: Use DSP's PID
 - To host edge: Use USP's PID
- DSAR flag

7.7.11.2 CXL VDMs

See [Section 3.1.11](#) for a list of VDMs that are used in the PBR fabric.

7.7.11.3 Single VH Events

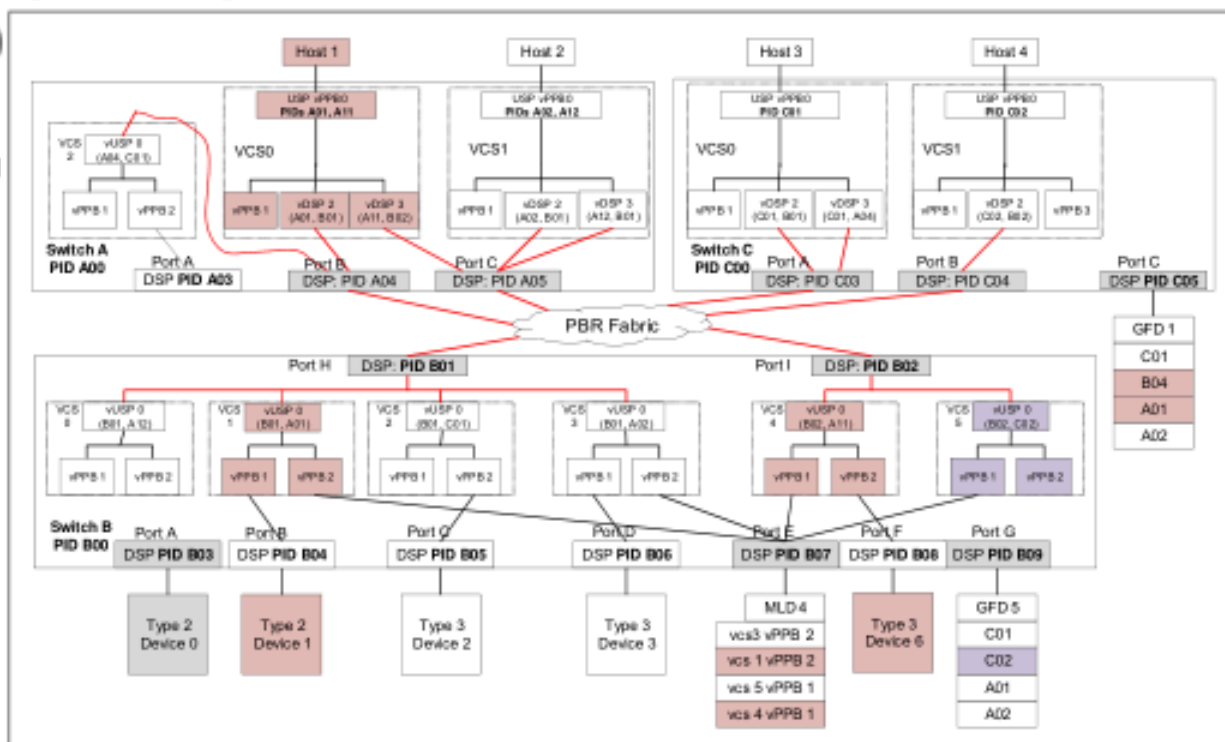
Events that are contained within a single VH should not affect other VHs that share an ISL.

PCIe visible events that are contained within a single VH include:

- Assert Reset
- Deassert Reset
- Link Up

[Figure 7-52](#) shows the virtual hierarchy from a Host 1 perspective (other hierarchies are grayed out). In Switch A, Host 1 finds only a single switch VCS 0. However, in Switch B, two switches VCS 1 and VCS 4 are in the Host 1 hierarchy. Switch B VCS 1 has vUSP 0 connected below Switch A VCS 0 vDSP 2, and Switch B VCS 4 has vUSP 0 below Switch A VCS 0 vDSP 3. Switch C has a GFD with that is accessible by Host 1 devices, but the GFD is not visible to the Host 1 PCIe hierarchy. See [Section 7.7.14](#) for more details on control of the GFD.

Figure 7-52. Single VH



7.7.11.3.1 Assert Reset VDM

Every PCIe hierarchy supports three levels of Conventional Reset:

- Fundamental cold reset (PERST#): Input pin
- Fundamental warm reset (PERST#): Input pin
- Hot reset due to Link Down, in-band hot reset, USP secondary bus reset, DSP secondary bus reset, or link disabled

CXL Fabric links support propagation of these resets. The ISL link state is not affected by any VH's Assert Reset or Assert PERST# VDM. Assertion of reset is accomplished using one of two different VDM opcodes:

- **Assert PERST#**: Used for fundamental reset assertion for that VH, Opcode 0
- **Assert Reset**: Used for hot reset assertion for that VH, Opcode 1

The separate PERST# message allows for fundamental reset functionality without the need for extra pins between switches.

Assert PERST# should be triggered whenever a VH has its input fundamental reset asserted on a Host ES. Assert Reset should be triggered whenever the Host ES:

- Receives a hot reset input
- Has a secondary bus reset on its USP
- Has a secondary bus reset on its VDSP
- Has a link disable on its vDSP

The Assert Reset VDMs all are sent from a vDSP to its paired vUSP. The VDM sent will have a PTH with:

- SPID = vDSP's host PID
- DPID = vUSP's FPort PID
- DSAR flag = 1

VDM header fields for Assert Reset VDMs:

- CXL VDM code of 80h
- PBR Opcode 0 or 1 indicates which Assert PERST# or Assert Reset message

It is expected that the Assert Reset VDM will reach a vUSP uniquely identified by the SPID and DPID at the destination switch.

A vDSP, upon sending Assert Reset VDM, will have its link state transition to Hot Reset.

A vUSP, upon receiving an Assert Reset VDM, will have its link state transition to Hot Reset. While in Hot Reset, all Port non-sticky registers and state machines that belong to the VH must return to their initialized state.

A vUSP, upon receiving an Assert PERST# VDM, shall have its link state transition to Hot Reset and also shall clear any sticky bits as outlined by PCIe Base Specification for PERST# behavior.

It is possible to send any number of Assert Reset VDMs or Assert PERST# VDMs.

In [Figure 7-53](#), if Host 1 asserts its PERST#, then both Switch A VCS 0 vDSP 2 and Switch A VCS 0 vDSP 3 shall issue an AssertPERST# VDM. The format of the PTH would be (SPID=A01, DPID=B01) for vDSP 2 and (SPID=A11, DPID=B02) for vDSP 3. If Host 1 instead asserted vDSP 2 secondary bus reset, then only vDSP 2 would send an AssertReset VDM with (SPID=A01, DPID=B01).

7.7.11.3.2 Deassert Reset VDM

A Deassert Reset VDM signals a release of reset and an exiting of the Hot Reset state to enter Detect for that VH. This VDM shall be sent from the Host Edge Switch due to a deassertion of the PERST# input resulting from an exit from Hot Reset.

If DSP is enabled the DPC trigger status must be cleared before a Deassert Reset VDM can be sent because DPC triggered prevents any TLPs from egressing that port.

Propagation of reset deassertion over an ISL is enabled via a Deassert Reset VDM, which is used for hot reset deassertion for that VH, Opcode 3.

A Deassert Reset VDM is used to instruct the vUSP to exit Hot Reset and enter Detect. The Deassert Reset VDM sent will have a PTH with:

- SPID = vDSP's host PID
- DPID = vUSP's FPort PID
- DSAR flag = 1

VDM header fields for Deassert Reset VDMs:

- CXL VDM code of 80h
- PBR Opcode 3

A vDSP, upon sending a Deassert Reset VDM, will have its link state transition from Hot Reset to Detect. A vUSP, upon receiving a Deassert Reset VDM, will have its link state transition from Hot Reset to Detect. If the link state is not in Hot Reset, a link state change will not occur.

The link for that VH will remain in Detect until the vUSP sends a Link Up VDM and the vDSP receives a Link Up VDM. If a Link Up VDM is not received within 10 ms, a subsequent Deassert Reset VDM shall be sent. This can repeat until 10 Deassert Reset VDMs have been sent. After a tenth Deassert Reset VDM is sent, if a Link Up VDM is still not received within 10 ms, the reset deassertion failed and the FM shall be notified.

In [Figure 7-53](#), if Host 1 clears the secondary bus reset in Switch A VCS 0 vDSP 2, then Switch A VCS 0 vDSP 2 would send a Deassert Reset VDM with (SPID=A01, DPID=B01). Switch B VCS 1 vUSP 0 would exit the hot reset state. As part of the exit from LTSSM Detect and due to the shared link nature of an ISL, Switch B VCS 1 vUSP 0 will bypass the PCIe LTSSM states of Polling and Configuration and transition the vDSP-to-vUSP link back to L0 (Link Up) by sending a Response Link Up VDM.

7.7.11.3.3 Link Up VDM

A Link Up VDM signals a transition to L0 active for that VH's link. The Link Up VDM is sent by a vUSP to its paired vDSP to convey a post-Detect state across the shared ISL.

The vUSP sends a Link Up VDM after receiving a Deassert Reset VDM. The vUSP can perform any required post-reset initialization before sending the Link Up VDM. The vUSP may take as long as it needs after Deassert Reset to send the Link Up VDM. Any number of Deassert Reset VDMs may be received by the vUSP; for each Deassert Reset VDM received, a Link Up VDM shall be sent.

The vUSP, after sending a Link Up VDM, shall have its link state transition to L0 from Detect. Polling and Configuration link states are bypassed by the Link Up VDM because the required TS1 and TS2 Ordered Sets cannot be sent over a shared ISL.

A vDSP, after receiving a Link Up VDM, shall have its link state transition to L0 from Detect. If not in Detect, there is no state change. Any number of Link Up VDMs may be received. Polling and Configuration link states are bypassed by the Link Up VDM, with the link directly transitioning from Detect to L0.

Neither a vDSP nor vUSP should ever have their link state reach Polling or Configuration state.

The VDM sent will have a PTH with:

- SPID = vUSP's FPort PID
- DPID = vDSP's host PID
- DSAR flag = 1

VDM header fields for LinkUp VDMs:

- CXL VDM code of 80h
- PBR Opcode 4

7.7.11.3.4 Dynamic vDSP-to-vUSP Bind

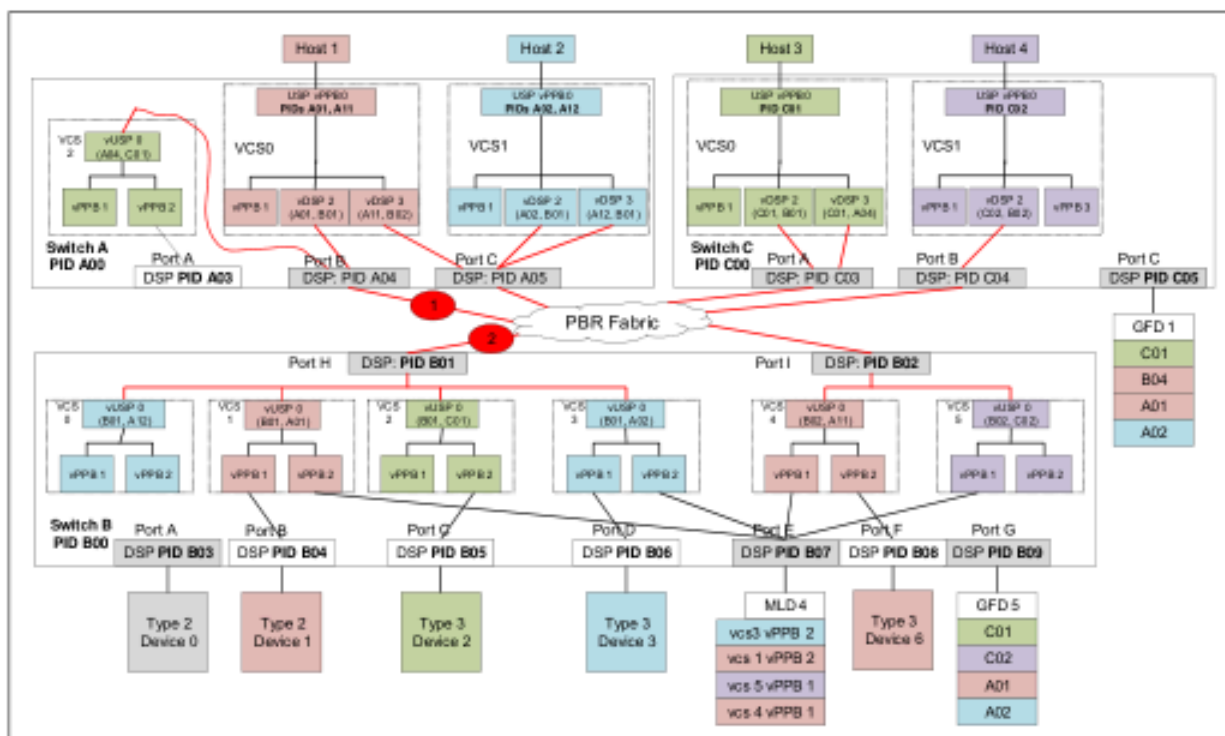
See [Section 7.7.12.3](#) for more details on the Configure PID Binding API sequence. After Configure PID Bind, the vDSP or vUSP shall be in a Hot Reset state. A vDSP may issue an Assert Reset VDM or a Deassert Reset VDM from the reset state, as dictated by its VH. A vUSP shall remain in Hot Reset until the vUSP receives a Deassert Reset VDM, upon which, after processing the necessary post-reset tasks, the vUSP will send a Link Up VDM.

7.7.11.4 Shared Link Events

Events that affect multiple VHS on the same link need to be reported to the FM. The FM shall take any necessary action.

The FM is required to keep an inventory for each ISL. Figure 7-53 shows how the link from Switch A Port B (indicated by an oval with 1) is shared by both a Host 1 hierarchy and a Host 3 hierarchy. Events on this link will affect both hierarchies. The oval with 2 is another shared link used by multiple hierarchies, of which only a Host 1 hierarchy is colored in but the ISL also includes Host 3 (VCS 2) and two hierarchies of Host 2 (VCS 0 and VCS 3).

Figure 7-53. Shared Link Events



7.7.11.4.1 Inter-Switch Link (ISL) Down

An ISL going down may affect one or more VHS.

A switch on each side of the ISL knows if the link had any issues. The fabric port's DPC is used to handle link issues. If DPC triggers, switch firmware will be notified. DPC may trigger due to Link Down or due to other reasons, such as software trigger; the net result is that the ISL will go down. Once the link goes down the switch reports the event to its primary FM. The FM is responsible for resolving the ISL Down event for all involved VHS.

The fabric port's DPC should remain triggered until switch firmware can resolve the side effects of an ISL Down event. When the FM has finished its resolution tasks, the FM will instruct the switch to clear the DPC trigger on the fabric port DSP. DPC trigger clear indicates resolution of the event and also allows the ISL to come back up.

The FM requires an inventory of users of an ISL to correctly resolve an ISL Down event. FM tasks for the resolution of an ISL Down event involves the following:

- Unbinding any affected VHs' vDSP
- Unbinding any affected VHs' vUSP
- Clearing any affected multi-path in a switch's RGT
- Clearing any affected GFD Access Vector in a switch's GAE

For example, if the link at Oval #1 in [Figure 7-53](#) breaks, Switch A and an unlabeled PBR fabric switch will both notify their primary FM. The FM will then unbind the following affected vDSPs and vUSPs:

- Switch A VCS 0 vDSP 2 and VCS 2 vUSP 0
- Switch B VCS 1 vUSP 0
- Switch C VCS 0 vDSP 2

As another example, if the link at Oval #2 in [Figure 7-53](#) breaks, Switch B and an unlabeled PBR fabric switch will both notify their primary FM. The FM will then unbind the following affected vDSPs and vUSPs:

- Switch A VCS 0 vDSP 2 and VCS 1 vDSP 3 and VCS 1 vDSP 2
- Switch B VCS 0 vUSP 0, VCS 1 vUSP 0, VCS 2 vUSP 0, and VCS 3 vUSP 0
- Switch C VCS 0 vDSP 2

In addition to the unbinding of the vDSP and vUSP pair affected by an ISL Down event, the RGT and GAE GFD access vectors may be updated by the FM. The RGT would be updated to avoid the path leading to the fault. The GFD Access Vector may be updated to remove a GFD that is no longer reachable.

7.7.11.5 Switch Reported Events

Some events are switch specific or are outside normal PCIe reporting methods and thus require switch-specific intervention. These include:

- Link Partner Info

7.7.11.5.1 Link Partner Info VDM

A Link Partner Info VDM is sent on all PBR links immediately after the InitFC process finishes for VC0. Each side of the link will send a Link Partner Info VDM at this time.

A Link Partner Info VDM also is sent whenever a payload field value is updated. Only the side of the link with an updated value needs to send the VDM.

This is a message with payload. For CXL 3.1, the payload is a fixed size of 16 DWORDs.

There are two types of PBR links: ISL and GFD. Both send the same Link Partner Info format but have a different value for the device type of the sender.

The Link Partner Info payload includes the following details about the sender of the VDM:

- 16B Link Partner ID: defined as the first 16 bytes of the Identify Output Payload as specified in [Table 8-50](#), for the hardware sourcing the Link Partner Info VDM Payload. Thus, this 16B string is a globally unique ID associated only with the sourcing hardware.
- 1B Physical Port ID: the ID number (port number) of the port sourcing (transmitting) the Link Partner Info VDM payload.
- 12bit PID (if FFFh, indicates sending port's PID is un-initialized)
- 4bit Device Type (0 = PBR switch, 1 = GFD, all other encodings are reserved)

- 1B Standard FC VC list
- 1B UIO FC VC list
- 16B FM Primary UUID. If this value has not been initialized, this value shall read all zeros.
- 16B FM Secondary UUID. If this value has not been initialized, this value shall read all zeros.

Table 7-110. Link Partner Info Payload

+3								+2								+1								+0								Byte Offset
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	
DevType[3:0]				PID[11:0]												Reserved[7:0]								PortID[7:0]								+0
Link Partner ID[127:0]																																+4
																																+8
																																+12
																																+16
Primary FM UUID[15:0]																UIO FC VC List[7:0]								Standard FC VC List[7:0]								+20
Primary FM UUID[111:16]																																+24
																																+28
																																+32
Secondary FM UUID[15:0]																Primary FM UUID[127:112]																+36
Secondary FM UUID[111:16]																																+40
																																+44
																																+48
Reserved																Secondary FM UUID[127:112]																+52
Reserved																																+56
																																+60

With multibyte fields, the least significant byte of the field starts with the lowest byte offset, and subsequent bytes are strictly increasing in significance. I.e., this is little endian format within each multibyte field as well as the overall payload.

The Link Partner Info VDM.PTH fields are as listed below. This VDM will terminate at the Receiver.

- SPID = Originator's (switch's/GFD's) PID, A value of FFFh indicates the sender's PID is un-initialized.
- DPID = FFFh (fixed value which indicates the receiving port is to process the VDM payload)
- DSAR flag = 1

VDM header fields for LinkPartnerInfo VDMs:

- Type 74h (Message with Data, terminate at Receiver)
- CXL VDM code of 90h
- PBR Opcode 0

A single message is sufficient to carry all the link info for CXL release 3.1.

7.7.11.6 PBR Link CCI Message Format and Transport Protocol

CCI commands are transported on PBR links as defined in [Section 7.6.3](#) and its associated binding specifications (see DSP0234, DSP0238, and DSP0281) with some notable caveats and clarifications:

- As with all .io traffic across PBR links, MCTP PCIe VDMs include a PTH whose SPID and DPID define the routing of the message
- PCIe enumeration is not required for ISL PPBs and GFDs
- GFDs do not implement a PCIe Physical Function
- "Requester ID" and "Target ID" fields in the VDM's TLP header are reserved because IDs are not assigned to many elements within the fabric (e.g., FM, ISL PPBs, Switch Management FW, GFDs, etc.)

7.7.12 PBR Fabric Management

7.7.12.1 Fabric Boot and Initialization

Much like as outlined for HBR switches in [Section 7.2.1](#), PBR switches may be initialized in one of three different ways:

- Statically
- FM boots before the host(s)
- FM and host boot simultaneously

7.7.12.1.1 Static Fabric Initialization

A static fabric deployment uses statically predefined configuration data to define the fabric configuration settings typically assigned dynamically by an FM.

Static Fabric Characteristics:

- No support for G-FAM or MLD
- No support for dynamic binding changes or DCD
- No FM is required, but may be needed for error handling
- At switch boot, all ports have a PID assigned, DRT and RGT tables are pre-populated, and EP and PID binding settings are predefined as defined by vendor-specific switch configuration data (e.g., configuration file in SPI Flash)
- Each VH is ready for enumeration when the host boots
- Hot-add and managed hot-remove are supported on Downstream Edge Ports

7.7.12.1.2 Fabric Manager Boots First

With this method, the FM configures the fabric binding relationships and access permissions before the host boots and enumerates its VH.

- FM boots while hosts are held in reset
- All attached ISLs and DSPs link up and, when negotiated in PBR mode, exchange the PBR Link Information VDM
- FM discovers fabric topology, claims ownership of all components under its management, and assign PIDs
- FM binds EPs to VCSs and configures GFDs
- FM configures GMV and VTV to enable G-FAM, GIM and Edge-to-edge P2P, as required when available

7.7.12.1.3 Fabric Manager and Host Boot Simultaneously

In the case where the switches, FM, and host boot at the same time:

- VCSs, PID assignment, GFD configuration, and bindings between Host ES to Downstream ES VCSs are statically defined
- Edge vPPBs within each VCS are unbound and presented to the host as Link Down
- Switch discovers downstream devices and presents them to the FM
- Host enumerates the VH and configures the DVSEC registers
- FM performs port binding to edge vPPBs
- Switch performs virtual to physical binding
- Each bound port results in a Presence Detect Change or Link State Change notification to the host
- For G-FAM access, FM updates GMV and VTV access vectors for hosts

7.7.12.2 PBR Fabric Discovery

To effectively manage a PBR fabric, the FM must understand the physical topology through a fabric discovery process. A typical fabric discovery may proceed as follows.

1. FM discovers the component to which it is directly connected and claims primary FM ownership.

Management of a PBR device requires that a primary FM is registered. A PBR device shall accept only the following commands from an FM that is not registered as the primary FM:

- **Identify**
- **Get Supported Logs**
- **Get Log**
- **Identify PBR Component**
- **Claim Ownership**

All other commands shall fail with “Unsupported Request”. A PBR device shall only advertise support for the CEL and the CEL shall only advertise the commands in the above list when the supported logs or CEL contents are queried by an FM that is not registered as the primary FM.

If the FM is connected to a switch, crawl out and discovery of the fabric continues.

2. FM explores all switch ports.

As primary FM, the switch capabilities and switch port status can be queried. The **Get Physical Port State** and **Get PBR Link Partner Info** commands provide information on the devices connected to each port.

PBR switches can determine the type of device present at the far end of a link after negotiation using the link state information provided in [Table 7-111](#).

Table 7-111. Far End Device Type Detection (Sheet 1 of 2)

Device Type	Negotiated Link Direction	Negotiated PBR-Enabled	Negotiated MLD-Enabled	Received “Link Partner Info” Type
Host	USP	N	N	N/A
PBR Switch	DSP-DSP Crosslink	Y	N	Switch